

Subsystem Specification

Mission computer

Take and geotag photos, relay MAVLink to the bank, and run the small boat-side watchdogs, without ever being needed for safety.

Document	BOATY-SSS-MCP
Issue	Issue E (for review)
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Status	For review by the project owner
Parents	SRS Issue G, ADD Issue G, ICD Issue G, FMEA Issue F
Content	14 allocated SRS requirements → 35 subsystem requirements (26 M, 7 S, 2 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	FMEA actions A-03, A-07, A-08, A-11: navigation monitor B7 (MCP-D22 to D25, D27) and box temperature (MCP-D26).	Claude, owner decisions (CR-02, FMEA actions)
C	29 September 2026	Simulator slice 2: B6 takes the FS-004 HOLD (MCP-D28) and the FS-001 slow RTL (MCP-D29); B7 takes FEN-006 (MCP-D30); B5 astern bursts by thrust command (MCP-D18, ICD IF-04 correction); MCP-D31 to D33 from integration. All SIM requirements now have passing scenarios (software/results).	Claude, owner decision (slice 2)
D	29 September 2026	SC-06 intermittent failure traced to B5's free test (astern drift counted as moving): MCP-D34 added (FMEA FM-58, A-29).	Claude
E	30 September 2026	SDR WP2: MCP-D30 far-outside trigger 10 → 8 m so the motors are off by FEN-006's 10 m (SC-28 failed at 10.2-10.6 m once the path planner's start pause changed which detector fired first). Event text now 'B7 FAR OUTSIDE FENCE: HOLD'. WP3: MCP-D35 position-jump HOLD (SC-20 showed a sustained 20 m GNSS offset took the boat 12 m outside the fence).	Claude

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Take and geotag photos, relay MAVLink to the bank, and run the small boat-side watchdogs, without ever being needed for safety.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Raspberry Pi Zero 2W, microSD, heatsink • 5 MP camera • Moisture sensor • OS image and services B1-B7 (software/boaty/mcp)	• Its 5 V supply (PWR) • Camera hood and mount (HUL) • Helm behaviour (HLM)

1.3 Breakdown

Item	Description
MCP-1 Computer	Pi Zero 2W, 32 GB A1/U3 microSD, heatsink
MCP-2 Camera	OV5647-class 5 MP, $\leq 62^\circ$ HFOV
MCP-3 Sensors	Moisture traces at the box low point; DS18B20 box temperature sensor
MCP-4 Software	B1 router, B2 camera, B3 photo server, B4 link watchdog, B5 weed-shedding, B6 health, B7 navigation monitor

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue D	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue C	Interface definitions (all listed in section 3 apply in full)
BOATY-FMEA-001 Issue B	Failure modes and actions carried into this issue

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-01 Boat-bank radio link	Mission Control	Conforms	\$4
IF-03 Photo & health service	Mission Control	Owner	\$4
IF-04 Companion link	Helm	Owner	\$4
IF-08 Mission computer power	Power	Conforms	\$6

4. Parent requirements

Primary (14): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
FEN-006	M	If the boat stays outside the inclusion fence for more than 30 s, or gets more than 10 m outside it, the motors shall...	MCP-D30
FS-004	M	Position loss: if the GNSS fix is lost, HDOP exceeds 2.5 or the position estimate is unhealthy for 3 s, the motors...	MCP-D15, MCP-D28
FS-006	S	Weed-shedding: after a stuck detection the boat shall try up to 3 short reverse bursts, then resume its previous mode...	MCP-D18, MCP-D34
FS-010	S	Water ingress: a moisture sensor in the electronics box shall trigger RTL and an alarm.	MCP-D03, MCP-D15
CAM-001	M	The boat shall carry a forward-looking camera that captures still images of at least 2 MP (M) and at least 5 MP (S).	MCP-D01
CAM-002	M	Capture modes shall be: interval (every 2 to 30 s), at photo points, and on command.	MCP-D10
CAM-003	M	Every photo shall be tagged with UTC time, position, heading and mission ID.	MCP-D11
CAM-004	M	Local storage shall hold at least 500 photos. Full storage shall not affect navigation or safety.	MCP-D06, MCP-D13
CAM-005	S	A duck (about 0.3 m long) at 5 m shall span at least 40 pixels in the image.	MCP-D02
CAM-007	S	After a mission, photos shall transfer to Mission Control automatically when in range, with 200 photos taking 5 min or...	MCP-D14, MCP-D16
CAM-008	C	Mission Control may show a low-rate live view from the camera.	MCP-D17
DET-003	C	An upgraded mission computer may detect waterfowl on board in real time and trigger 'pause and look' (hold and take a...	MCP-D20
DET-004	M	If DET-003 is implemented, 'pause and look' shall hold position and shall never steer towards the detected animal.	MCP-D21
PWR-010	M	Electronics in the sealed box shall run for 60 min at 30 °C ambient in direct sun without fault or throttling.	MCP-D04, MCP-D26

Supporting (17): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

PRE-008, FS-001, FS-002, FS-003, FS-005, FS-007, CAM-006, COM-001, COM-002, COM-003, COM-004, COM-006, PWR-005, SWE-004, SWE-008, LOG-004, ENV-005

5. Subsystem requirements

Requirements are numbered MCP-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Hardware

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D01	The mission computer shall be a Pi Zero 2W with a 32 GB A1/U3 microSD and a 5 MP camera.	M	I	BENCH	CAM-001, DD-03
MCP-D02	Camera horizontal field of view shall be $\leq 62^\circ$, giving ≥ 120 px across a 0.3 m duck at 5 m (at 2592 px width).	S	A	BENCH	CAM-005
MCP-D03	A moisture sensor (exposed traces at the box's lowest point) shall be read by GPIO.	S	T	BENCH	FS-010
MCP-D04	The SoC shall have a heatsink, and 60 min at 30 °C in the sealed box in sun shall give no throttling (throttled flag = 0).	M	T	BENCH	PWR-010, V-10
MCP-D05	Average power ≤ 2.0 W, peak ≤ 3.0 W.	S	T	BENCH	PWR-011

5.2 Platform software

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D06	The OS shall be Raspberry Pi OS Lite 64-bit with a read-only root filesystem and a separate data partition.	M	T	BENCH	CAM-004, IF-08
MCP-D07	Services B1-B7 shall be systemd units that restart automatically and are running ≤ 45 s after power-on.	M	T	BENCH	IF-04, IF-03
MCP-D08	The clock shall be set from helm GNSS time at start (no internet on the boat).	S	T	BENCH	SWE-008
MCP-D09	SSH shall be key-only, and the firewall per IF-01.	M	I	BENCH	COM-006, IF-01

5.3 Camera and photos (B2, B3)

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D10	B2 shall capture on interval (2-30 s during AUTO), at photo points (burst_n within 0.5 s of arrival) and on command.	M	T	SIM	CAM-002, IF-03
MCP-D11	Each photo shall carry UTC time, lat/lon/heading (≤ 0.5 s old, else null) and mission ID, in EXIF and the index.	M	T	SIM	CAM-003
MCP-D12	B4 (link watchdog): in AUTO with no system-255 heartbeat for 60 s $\rightarrow$ RTL. In HOLD after a MANUAL link loss, at 10 s $\rightarrow$ RTL.	M	S	SIM	FS-003, FS-002
MCP-D13	≥ 5 GB shall be reserved for photos (≥ 500 at ≤ 2.5 MB). Capture stops below 500 MB free. Navigation is unaffected.	M	T	BENCH	CAM-004
MCP-D14	B3 shall implement IF-03, with thumbnails made at capture time.	M	T	BENCH	IF-03, CAM-007
MCP-D15	B6 (health): moisture or box > 60 °C $\rightarrow$ RTL within 2 s plus STATUSTEXT, unless the position is unhealthy (then HOLD wins, FS-011). After a position-loss HOLD, RTL once the position has been healthy for 10 s.	M	S	SIM	FS-010, FS-004, FS-011
MCP-D16	Photo sync throughput ≥ 11 Mbit/s with the boat ≤ 10 m from the bank (200 photos ≤ 5 min).	S	T	POOL	CAM-007
MCP-D17	Optional live view: 320 $\times$ 240 MJPEG at 2 fps when enabled, paused during sync.	C	D	POOL	CAM-008

5.4 Weed-shedding and command filter (B5)

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D18	B5 shall act only on a stuck event (helm crash check or B7). Up to 3 GUIDED astern bursts (≤ 0.5 m/s, ≤ 2 s, commands at 10 Hz), then resume the previous mode; keep it if the boat is free (MCP-D34), else HOLD + alarm. Bursts use SET_ATTITUDE_TARGET thrust with zero yaw rate: a negative velocity target makes Rover turn round instead (ICD IF-04, Issue E).	S	S	SIM	FS-006, V-11
MCP-D19	All services shall send MAVLink only through a shared client with a default-deny filter (IF-04 list, per-service policy). Proven by fuzzing every forbidden command from every service in every helm mode (SC-30).	M	T	SIM	SAF-003, IF-04

5.5 Future: on-board duck spotting

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D20	An upgrade may run an on-device bird detector at ≥ 2 fps that requests 'pause and look'.	C	T	POOL	DET-003
MCP-D21	'Pause and look' shall only request LOITER (≤ 20 s) and then resume. It shall never steer towards the detection. Applies only if MCP-D20 is implemented.	M	S	SIM	DET-004

5.6 Navigation monitor B7 and box temperature (Issue B)

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D22	First-motion heading check: in the first 10 s of AUTO, RTL or STEERING, once above 0.3 m/s, a GNSS-course vs heading difference $> 45^\circ$ for 3 s shall request HOLD and raise an alarm.	M	S	SIM	FS-013, A-03, FM-05, FM-18
MCP-D23	Second stuck detector: throttle $\geq 50\%$ and progress along the active leg < 0.1 m/s for 10 s in AUTO or RTL shall request HOLD and raise a stuck event, which B5 acts on.	S	S	SIM	FS-005, A-07, FM-16
MCP-D24	Divergence watchdog: in AUTO or RTL, cross-track error > 10 m or heading error to target $> 60^\circ$ for 20 s shall request HOLD and raise an alarm.	M	S	SIM	FS-013, A-08, FM-17
MCP-D25	B7 shall only request HOLD, through the IF-04 filter. It shall never resume, arm or change mode otherwise. Resuming is an adult decision at Mission Control.	M	T	SIM	SAF-003, IF-04
MCP-D26	A DS18B20 at the top of the box interior shall be read at 1 Hz. Above 60°C , B6 requests RTL and raises an alarm. The value is reported as box_temp_c (IF-03).	M	T	BENCH	PWR-010, IF-03, A-11, FM-23
MCP-D27	B7 thresholds shall be configuration, version-controlled, and exercised by SC-05, SC-29 and SC-38.	M	I	SIM	SAF-007, A-03, A-07, A-08

5.7 Gaps the helm cannot close (ADD DD-21)

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D28	B6 shall request HOLD when the position has been unhealthy (no 3D fix, HDOP > 2.5 or EKF unhealthy) for 1 s as the helm reports it. The helm takes ~ 1.5 s to report a loss, so the motors stop within FS-004's 3 s of the loss itself (2.5-2.7 s in SITL). The native EKF failsafe alone took 9 s.	M	T	SIM	FS-004, DD-21
MCP-D29	At critical battery ($\leq 15\%$ or the helm's critical alarm), B6 shall reduce the RTL speed to 0.6 m/s with DO_CHANGE_SPEED and raise an alarm. It may only reduce speed, and only in RTL.	M	T	SIM	FS-001, V-15, DD-21

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D30	B7 shall request HOLD when the boat has been outside the fence for 30 s or is more than 8 m outside it (distance computed from the fence read back from the helm), in any powered mode. The 8 m trigger leaves room for detection latency so the motors are off by FEN-006's 10 m (SC-28, SDR WP2).	M	T	SIM	FEN-006, V-14, DD-21, FM-14

5.8 Found in integration (Issue C)

ID	Requirement	Pri	Ver	Stage	Trace
MCP-D31	B7's stuck and divergence checks shall be suspended within 5 m of the active target: when station-keeping at home, progress along a leg is meaningless and wind can push throttle past 50%.	M	T	SIM	FS-005, A-07
MCP-D32	A service shall not send mode-dependent commands until the helm's heartbeat reports that mode (the filter checks against the reported mode). B5 waits for GUIDED before its first burst.	M	T	SIM	SAF-003, IF-04
MCP-D33	All MAVLink shall be version 2 from the first packet (IF-02). MAVLink 1 silently drops extension fields such as mission_type.	M	I	SIM	IF-02
MCP-D34	B5 shall judge the boat free only on forward speed (along the heading) above 0.2 m/s held for 1 s within 8 s of resuming: the burst leaves the boat drifting astern, which ground speed counts as moving (SC-06). More than 3 episodes in 120 s means HOLD and a 'repeatedly stuck' alarm (a dead motor looks like weed). A refused or unconfirmed GUIDED switch is retried once, then reported as 'no control', never as 'still stuck'. Each watch is logged.	M	T	SIM	FS-006, FM-58
MCP-D35	B6 shall request HOLD when the helm's position jumps further than the boat could have moved (by more than 5 m beyond speed + 1 m/s times the interval), after at least 5 s of healthy position. The hold is latched: no automatic RTL, since home may be wrong too. An adult resumes or brings the boat home. A reset just after a position loss is expected and ignored (SC-04).	M	T	SIM	FS-013, FM-02, A-30

6. Timing

Event	Response	Budget
MISSION_ITEM_REACHED (photo point)	First photo of the burst	≤ 0.5 s
Interval tick	Photo captured	± 0.2 s
Moisture detected	RTL requested	≤ 2 s
No system-255 heartbeat in AUTO	RTL requested	$60 \text{ s} \pm 1 \text{ s}$
Helm stuck event	First astern burst	≤ 1 s
Power-on	All services running	≤ 45 s
POST /v1/shutdown	Filesystems synced, halt	≤ 10 s

7. Design constraints and budgets

- ADD DD-03 / T4: Pi Zero 2W (Python)
- ADD section 4.3 rule: may request only safer states; B5 bounded GUIDED
- SWE-001: Python services

Budget	Allocation
Mass allocation	≤ 35 g (estimate 30 g)
Cost allocation	£29 (Pi £15, camera £8, SD £4, temperature sensor £2); the flight controller's SD card is in HLM
Power	≤ 2.0 W average, ≤ 3.0 W peak

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	11	-	2	-	7	20
BENCH	9	-	2	1	-	12
POOL	2	1	-	-	-	3
LAKE	-	-	-	-	-	0

Ref	Open item
V-10	Thermal in the sealed box
V-11	Answered: the helm stops 3.9 s after the last target (3 s timeout + deceleration); accepted
B1	The simulator uses a Python router with mavlink-router's topology; the boat runs mavlink-router itself (check on rig L1)